Sales Performance & Discount Impact Analysis

Uncovering insights from **12,575 transactions** across product categories – analyzing spending patterns, customer segments, product preferences, and discount impact to guide strategic business decisions.

Dataset at a Glance

12,575

Transactions

11

Columns

25

Customers

8

Categories

Key Features

- **Transaction Details:** ID, Date
- **Customer Info:** ID, Location, Payment
- **Purchase Details:** Item, Category, Price, Quantity, Total Spent
- **Behavior:** Discount Applied

Missing data handled via imputation: Item → "Unknown", Discount → "No", numeric fields → mean/median by category.

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Data Preparation in Python

01

Load & Explore

Imported with **Pandas**; used `df.info()` and `.describe()` for structure and summary stats.

02

Clean & Standardize

Imputed missing values, renamed columns to **snake_case**, and verified price × quantity = total spent.

03

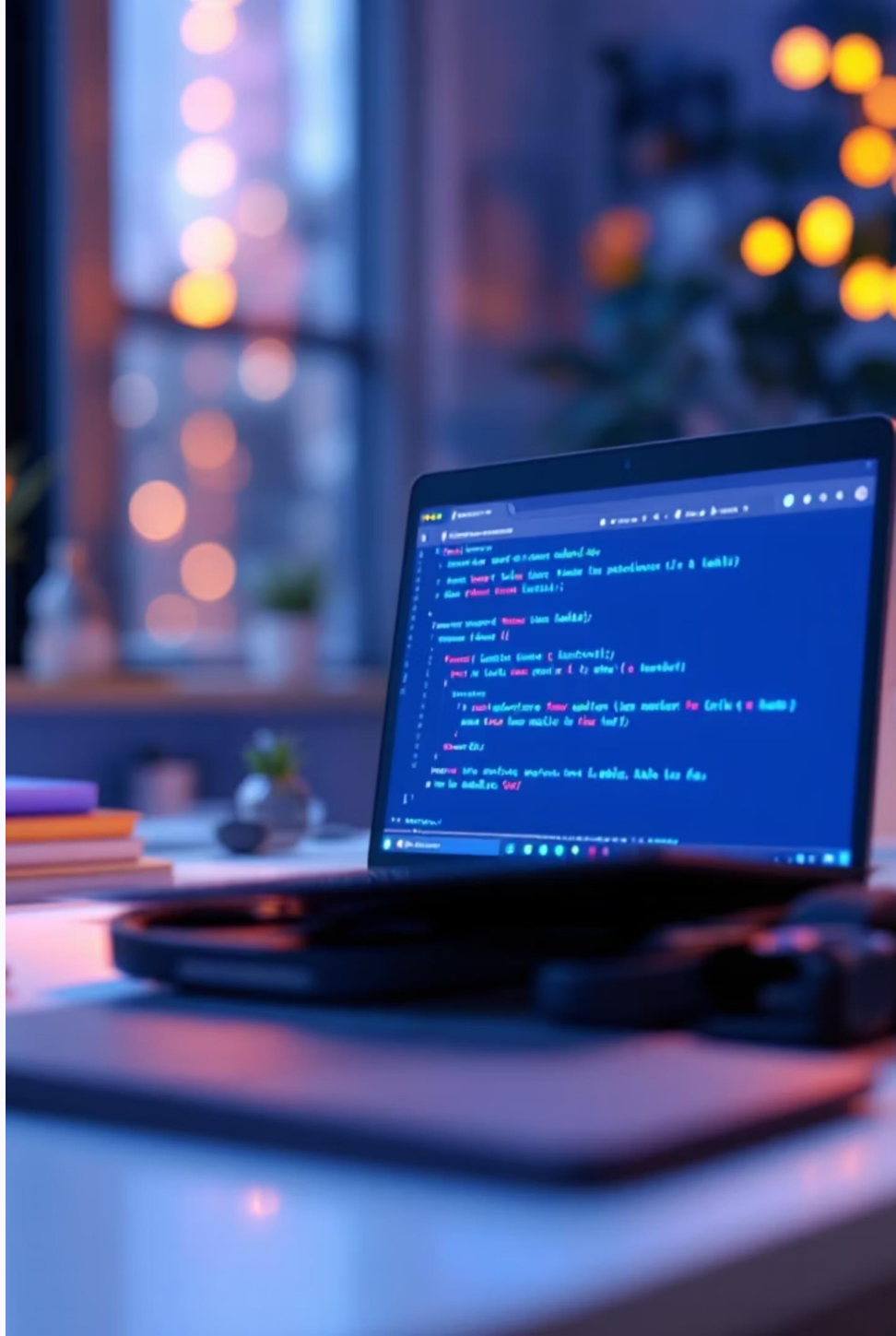
Feature Engineering

Extracted **year, month, and day** from transaction date for time-series analysis.

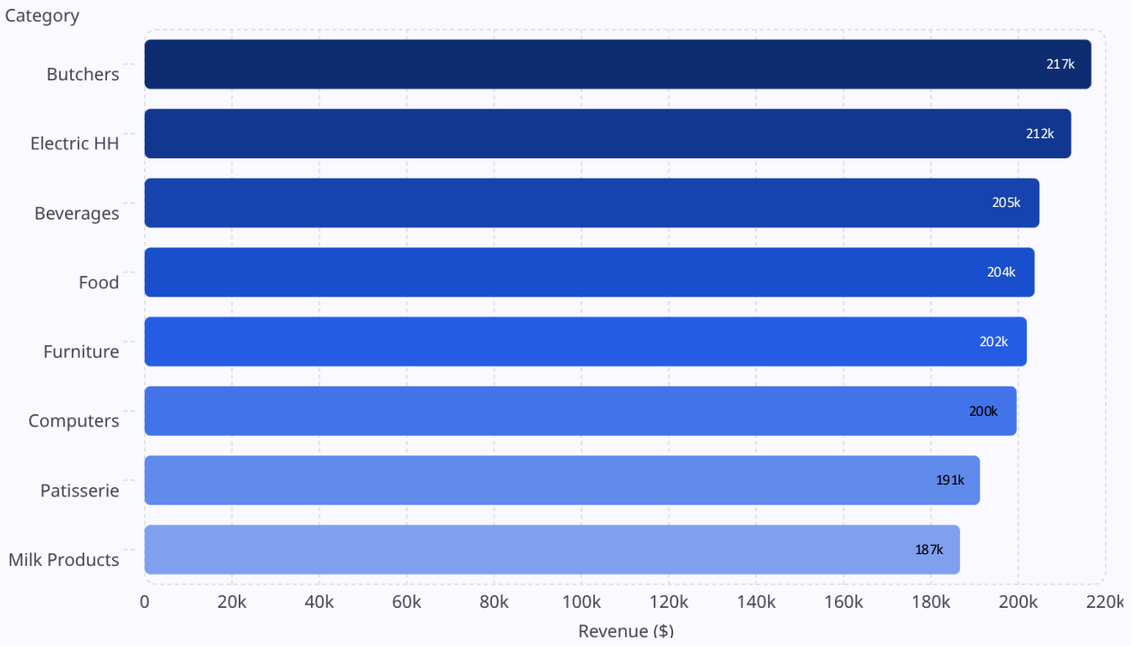
04

Database Integration

Loaded cleaned DataFrame into **MySQL** for structured SQL analysis.



Revenue & Sales by Category



Key Takeaways

Butchers leads revenue at $216,704, while **Milk Products** trails at $186,644. All categories contribute **~11-13%** each – a remarkably balanced portfolio.

Food leads in total sales volume (**8,873 units**), while Patisserie has the lowest at 8,465 – suggesting pricing differences drive the revenue gap more than volume.

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Online vs. In-Store Performance

Channel Insights

Online edges ahead in both revenue ($822K) and sales (35,416 units) vs. In-store ($794K, 34,412 units).

Average spend is nearly identical: **Online $129.38** vs. **In-store $127.74** – both channels are equally valuable.

Category split: Electronics & Beverages perform better Online; Furniture & Food lead In-store.

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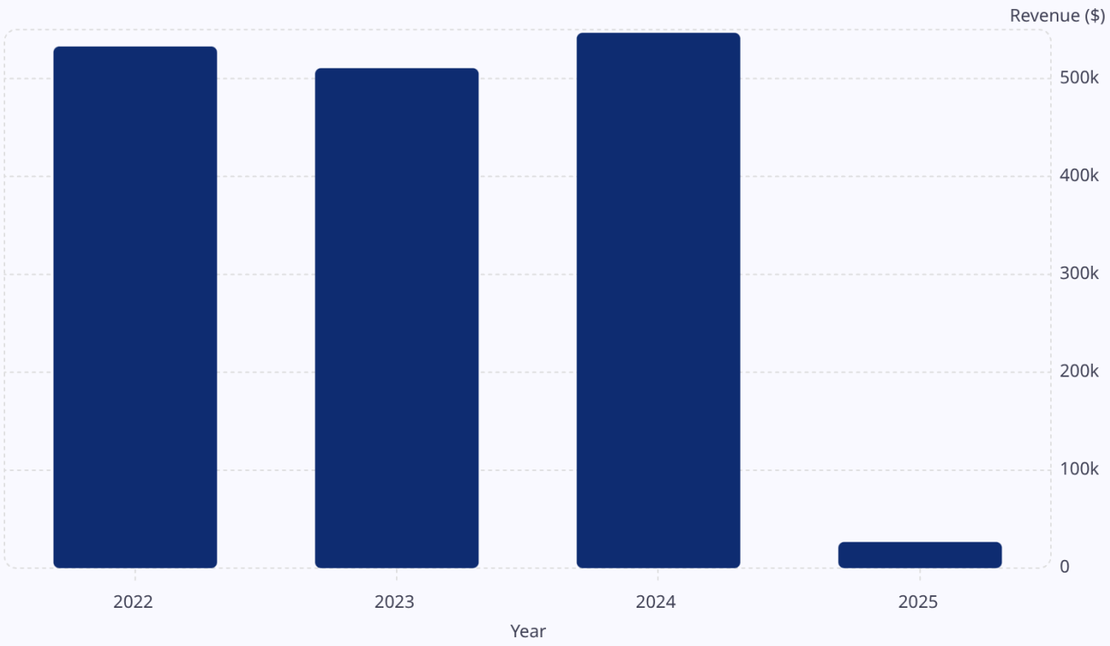
Revenue Over Time



Year-on-Year Trend

2024 recorded the highest revenue at $546,700, recovering from a dip in 2023 ($510K).

2025 shows only $26,678 – reflecting a partial year of data, not a decline.

Replicating 2024 strategies is a key recommendation for driving 2025 growth.

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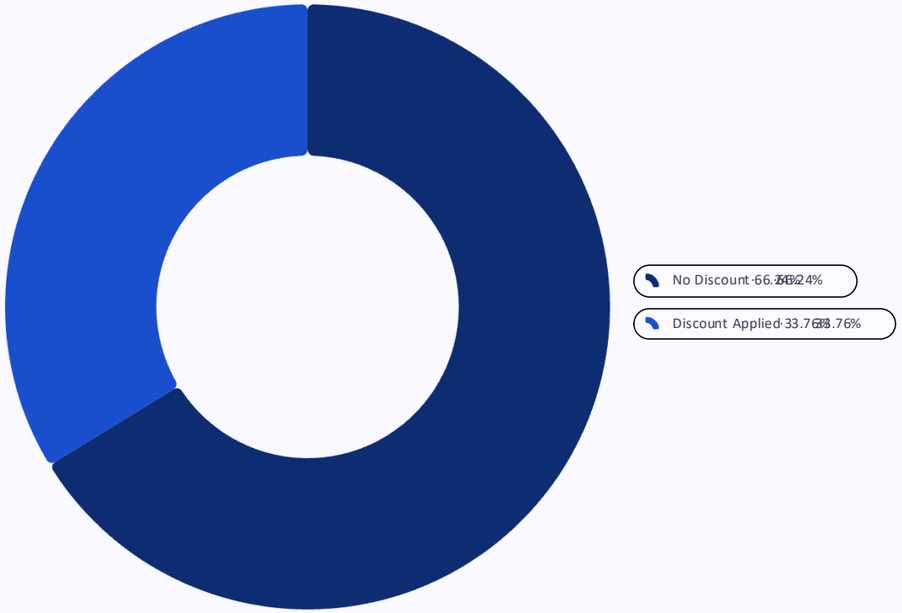
Discount Impact on Revenue



Discounts Don't Drive Revenue

Non-discounted orders generate **66.2% of total revenue** ($1.07M) vs. discounted orders at 33.8% ($545K).

- No-discount orders: **8,356** transactions
- Discounted orders: **4,219** transactions
- Average spend is nearly equal across both groups (~$129)

Broad discounts may not be necessary – strategic, targeted discounting is recommended.

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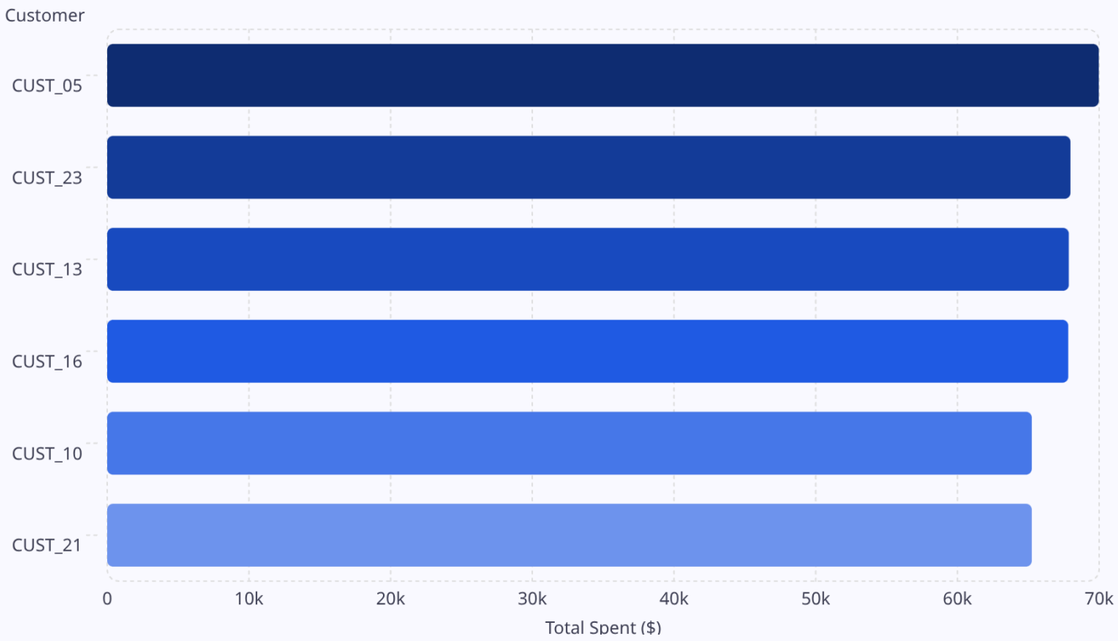
Customer Behavior & Top Spenders



High-Value Customers

CUST_05 is the top spender at $69,974 and also the most frequent buyer with **544 transactions.**

The top 25 customers show relatively balanced purchase frequency (465 - 544 transactions), indicating a loyal, active customer base.

A loyalty program targeting top spenders and frequent buyers could significantly boost retention.

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Retail Scan Dashboard

25
Customers

33.6%
Discount %

13K
Total Orders

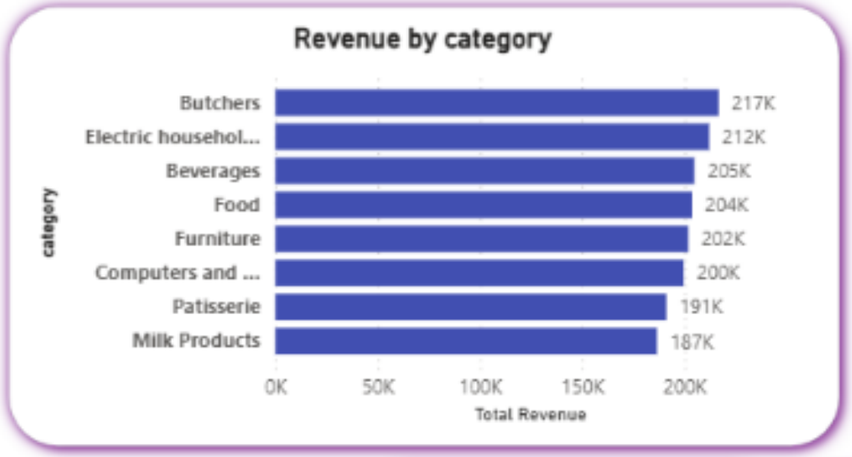
1.62M
Total Revenue

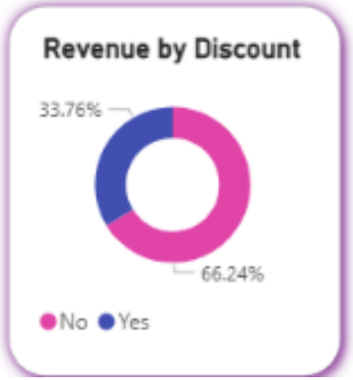
Discount Applied
- [ ] No
- [ ] Yes

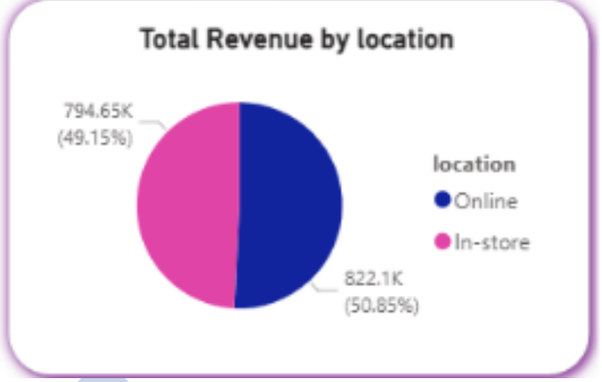
Location
- [ ] In-store
- [ ] Online

Category
All

Year, Month
All







Business Recommendations

📈 Invest in Top Categories

Prioritize **Butchers, Electronics & Beverages** to maximize revenue growth.

🎯 Strategic Discounting

Non-discounted orders drive **66.2% of revenue** – reduce broad discounts, apply them selectively.

🛒 Channel-Specific Marketing

Promote **Electronics Online**; focus **Furniture & Food** campaigns In-store.

🏆 Customer Loyalty Program

Reward top spenders and frequent buyers to drive repeat purchases.

🔁 Replicate 2024 Success

2024 was the peak revenue year – analyze and replicate those strategies for 2025.

📊 Improve Data Quality

Reduce missing values by implementing better data capture at point of transaction.

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